

Single Phase Synchronous motor

It works on the principle of **magnetic reluctance**

A magnetic material always tends to move in such a way that **reluctance of the magnetic path is minimum**.

In other words, the rotor aligns itself with the stator magnetic field to minimize magnetic reluctance.

Construction

Part	Description
Stator	Same as that of a single-phase induction motor, usually having a main and auxiliary winding (for starting).
Rotor	A salient-pole type rotor made of ferromagnetic material with slots to create varying reluctance — no winding, no commutator.

Working

1. Starting (As Induction Motor)

- When supply is given, the stator produces a **rotating magnetic field (RMF)** (using auxiliary winding + capacitor).
- The rotor starts as a **single-phase induction motor**, accelerating toward synchronous speed.

2. Pull-in to Synchronism

- As it nears synchronous speed, the rotor magnetic saliencies **lock in with** the stator field position — because this gives **minimum reluctance**.
- The rotor is then **pulled into synchronism** and runs at synchronous speed.

3. Running (Synchronous Mode)

- Once synchronized, it continues to run at synchronous speed, even with load variation (until torque exceeds pull-out torque).



Characteristics

Parameter	Description
Starting torque	Low
Running torque	Moderate
Efficiency	High
Speed	Constant (synchronous)
Power factor	Better than hysteresis motor
Applications	Electric clocks, timing devices, recorders, signal devices



Advantages

- Simple and robust construction
- Constant speed
- Better efficiency than hysteresis motor



Disadvantages

- Low starting torque
- Needs auxiliary means for starting

II. HYSTERESIS MOTOR

Principle

It works on the principle of **hysteresis loss** in a magnetic material.

The rotor tends to align with the rotating magnetic field because of **hysteresis lag** between the stator field and the rotor magnetization.

This produces **torque due to hysteresis effect**.

Construction

Part	Description
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Stator	Same as single-phase induction motor (main + auxiliary winding).
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Rotor	Made of hardened steel or cobalt steel cylinder, smooth (no slots or winding) , magnetically hard material with a wide hysteresis loop .
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Working

1. When supply is given, the **rotating magnetic field** is produced in the stator.
2. The **rotor gets magnetized** by this field — due to hysteresis, its magnetization **lags behind** the stator field.
3. This lag produces **torque** that causes the rotor to rotate in the **same direction** as the RMF.
4. The rotor **gradually accelerates** and **locks into synchronism**.
5. Once synchronized, it runs at **synchronous speed**, and torque is maintained by **hysteresis and eddy-current effects**.



Characteristics

Parameter	Description
Starting torque	High (better than reluctance motor)
Running torque	Constant (independent of speed)
Speed	Constant (synchronous)
Noise	Very quiet operation
Efficiency	Low (due to hysteresis losses)
Power factor	Poor
Applications	Clocks, electric timers, sound recorders, precision instruments



Advantages

- **Self-starting** (no need for auxiliary means)
- **Constant torque and speed**
- **Quiet operation**
- **Smooth running** (no slot ripple)

✗ Disadvantages

- Low efficiency
 - Expensive (hard magnetic rotor material)
 - Limited to small power ratings
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3. Comparison Table

Feature	Reluctance Motor	Hysteresis Motor
Rotor	Salient poles with variable reluctance	Smooth cylindrical hard steel
Starting Torque	Low	High
Starting Method	Needs auxiliary winding	Self-starting
Principle	Minimum reluctance	Hysteresis effect
Speed	Synchronous	Synchronous
Efficiency	High	Low
Noise	Slightly higher	Very quiet

Application	Clocks, timing devices	Precision instruments, recorders
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In Short

Motor	Principle	Main Advantage
Reluctance	Rotor aligns to minimize magnetic reluctance	Simple, efficient
Hysteresis	Rotor torque due to hysteresis lag	Self-starting, smooth & quiet

If the number of salient poles on the rotor is sub multiple of the stator poles, the motor will operate at a constant speed which is a sub-multiple of the synchronous speed. This is called a sub-synchronous reluctance motor